

Data Engineering

Spring 26 – DS471

Jan 23, 2026

Lecture 3

Data Science and Data Engineering

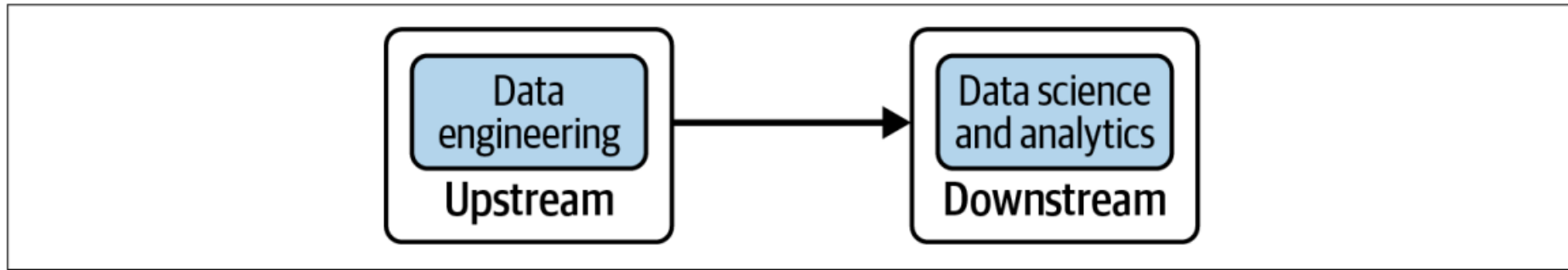


Figure 1-4. Data engineering sits upstream from data science

Data Science and Data Engineering

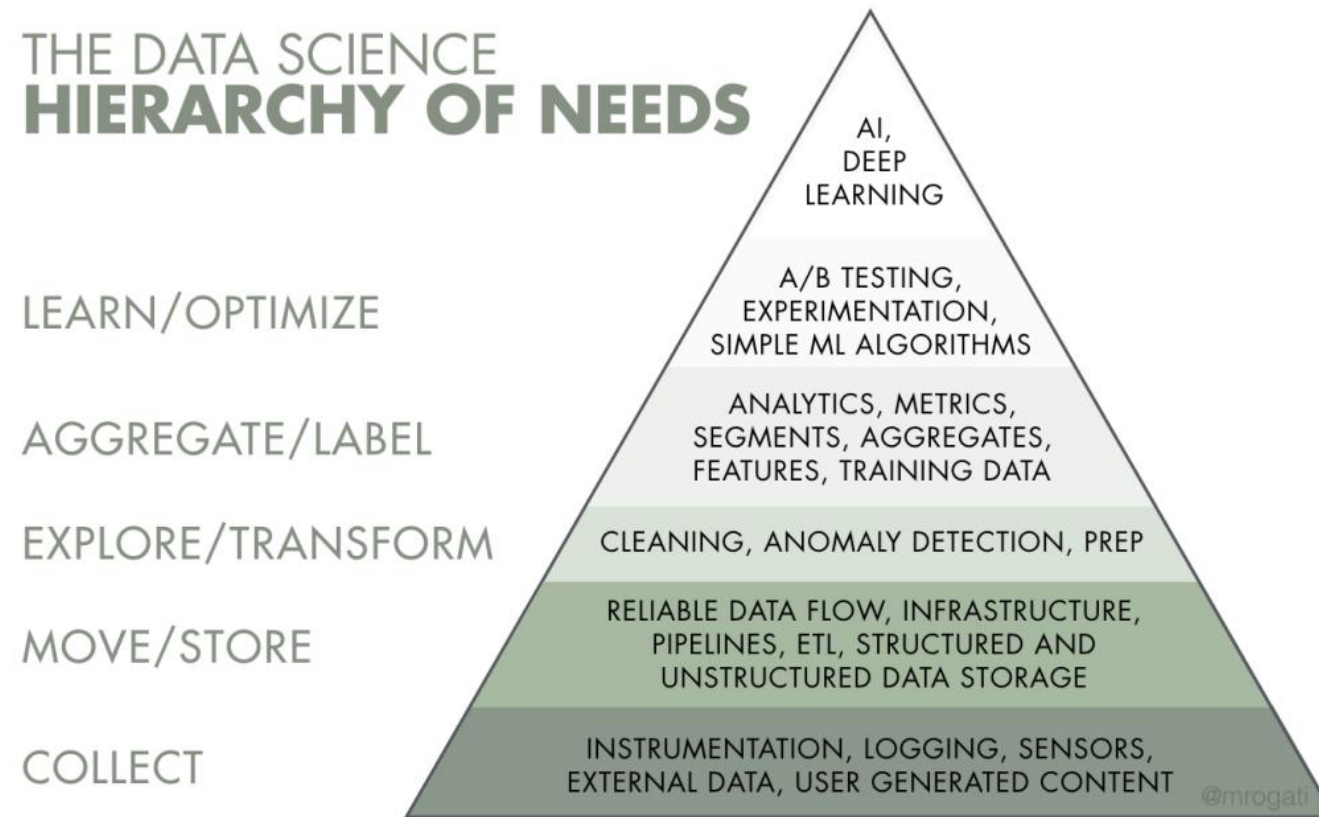


Figure 1-5. *The Data Science Hierarchy of Needs*

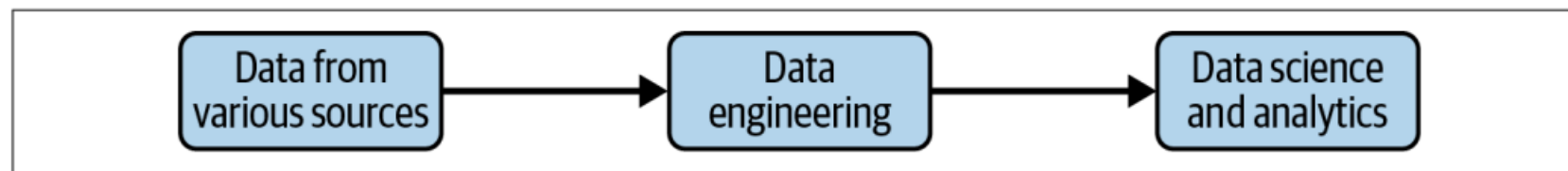


Figure 1-6. *A data engineer gets data and provides value from the data*

Data Maturity and the Data Engineer

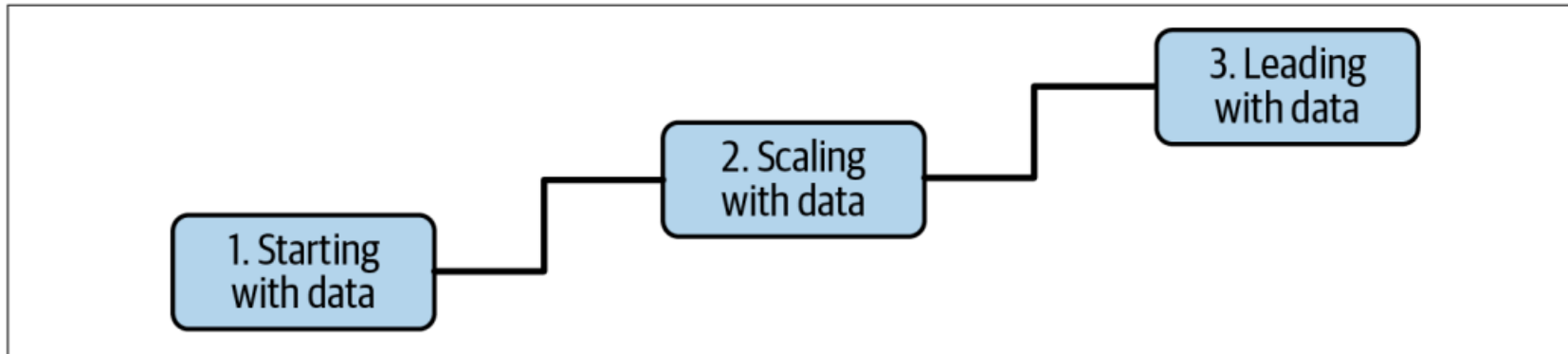
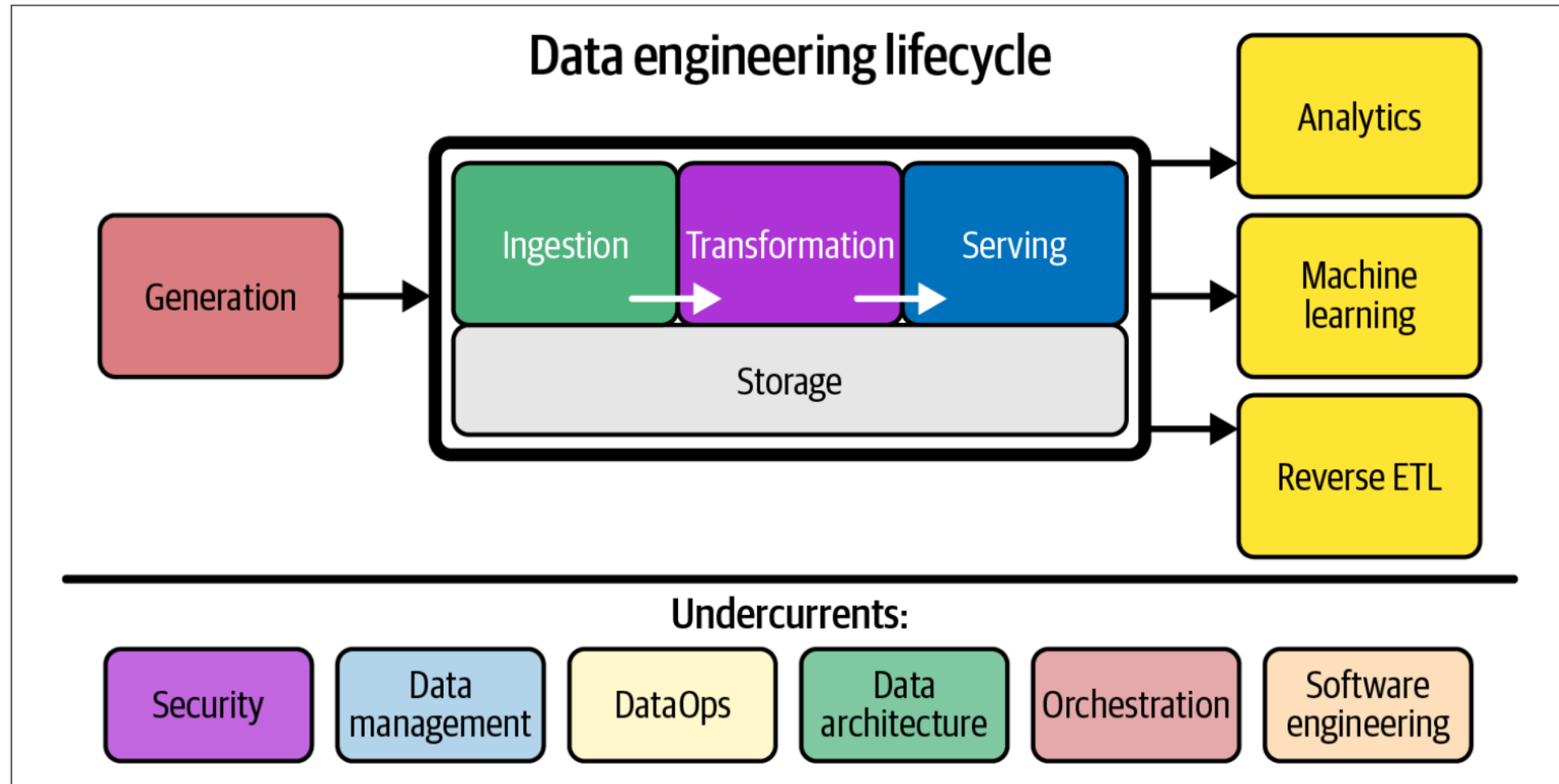


Figure 1-8. Our simplified data maturity model for a company

Business Responsibilities of data engineer

- Know how to communicate with nontechnical and technical people.
- Understand how to scope and gather business and product requirements.
- Understand the cultural foundations of Agile, DevOps, and DataOps.
- Control costs.
- Learn continuously

Technical Responsibilities of data engineer



SQL Python Java or Scala bash and more tools

The Continuum of Data Engineering Roles, from A to B

Type A Data Engineer

Abstraction

Type A data engineers manage the data engineering lifecycle mainly by using entirely off-the-shelf products, managed services, and tools.

Type B Data Engineer

Build

Type B data engineers build data tools and systems that scale and leverage a company's core competency and competitive advantage.

Data Engineers inside an organization

Internal-Facing Versus External-Facing Data Engineers

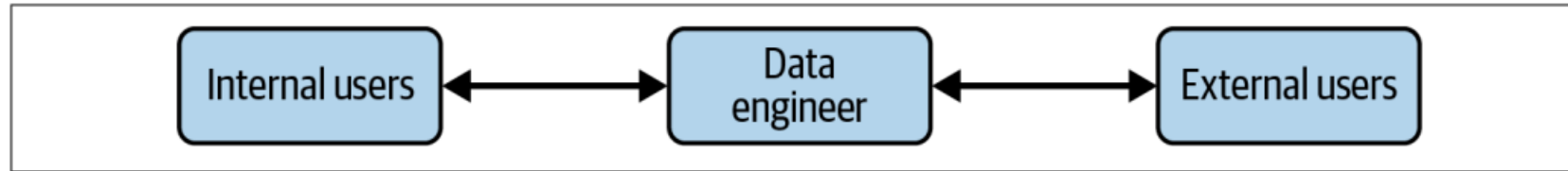


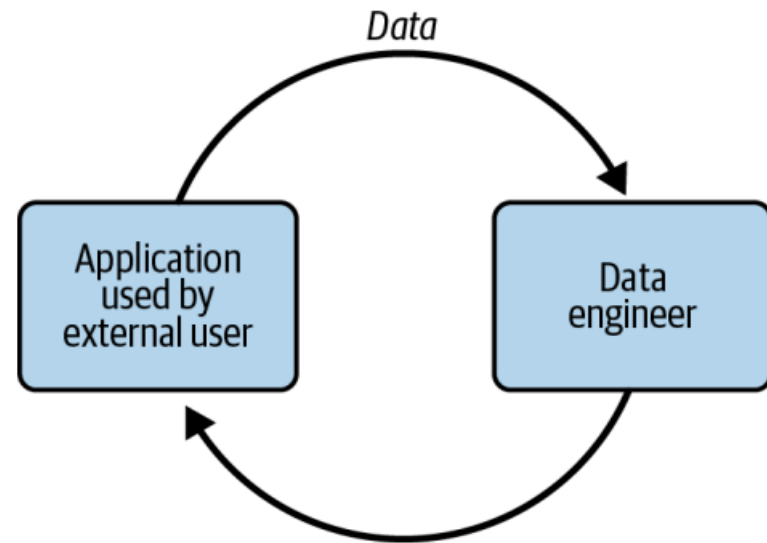
Figure 1-9. The directions a data engineer faces

- An **external-facing** data engineer typically aligns with the users of external-facing applications, such as social media apps, Internet of Things (IoT) devices, and e-commerce platforms
- An **internal-facing** data engineer typically focuses on activities crucial to the needs of the business and internal stakeholders. E.g. creating and maintaining data pipelines and data warehouses for BI dashboards, reports, business processes, data science, and ML models

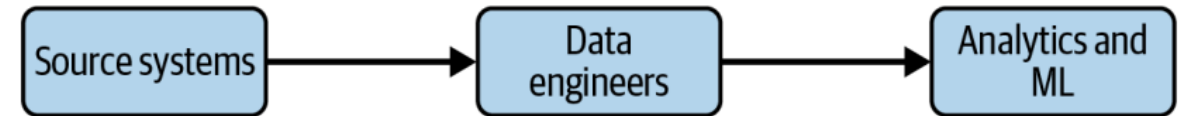
Data Engineers inside an organization

Internal-Facing Versus External-Facing Data Engineers

- **External-facing** data engineer systems



- **Internal-facing** data engineer systems



Data Engineering and other technical roles

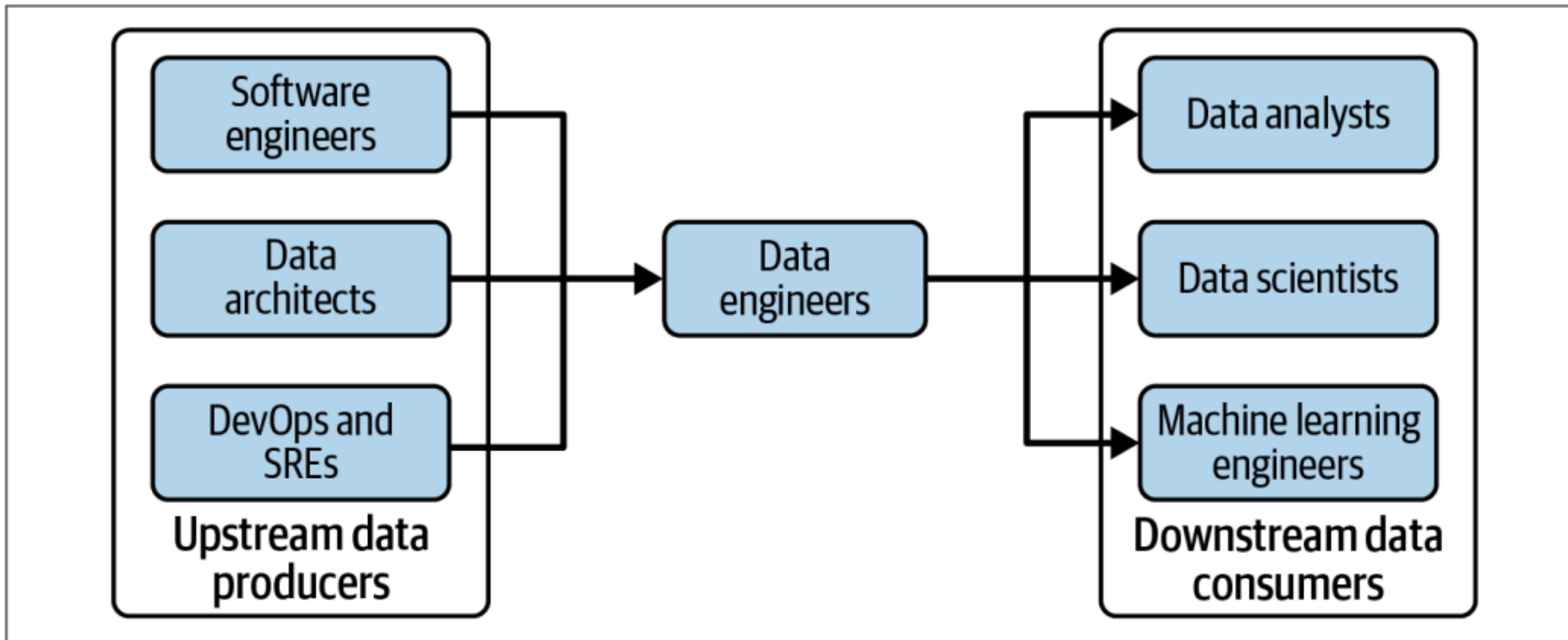


Figure 1-12. Key technical stakeholders of data engineering